LABORATORY PROGRAM – 2

Perform the following DB operations using

Cassandra Questions:

a) Create a keyspace by name Employee
b) Create a column family by name
- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch
d) Update Employee name and Department of Emp-Id 121
e) Sort the details of Employee records based on salary
f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
g) Update the altered table to add project names.
h) Create a TTL of 15 seconds to display the values of Employees.

```
cqlsh> CREATE KEYSPACE Employee
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

cqlsh>
cqlsh> USE Employee;
cqlsh:employee> CREATE TABLE Employee_Info (
            ...     Emp_Id INT PRIMARY KEY,
            ...     Emp_Name TEXT,
            ...     Designation TEXT,
            ...     Date_of_Joining DATE,
            ...     Salary INT,
            ...     Dept_Name TEXT
            ... );
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (121, 'Manoj', 'Engineer', '2022-01-10', 50000, 'CSE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (122, 'Rahul', 'Manager', '2021-05-15', 70000, 'ISE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (123, 'Anjali', 'Analyst', '2023-03-20', 45000, 'ECE');
            ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info
            ... SET Emp_Name = 'Manoj Kumar', Dept_Name = 'AI'
            ... WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info ORDER BY Salary DESC ALLOW FILTERING;
InvalidRequest: Error from server: code=2200 [Invalid query] message="ORDER BY is only supported when the partition key is restricted b
y an EQ or an IN."
cqlsh:employee> ALTER TABLE Employee_Info
            ... ADD Projects SET<TEXT>;
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'Hadoop', 'Spark'}
            ... WHERE Emp_Id = 121;
cqlsh:employee>
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'ML', 'Data Mining'}
            ... WHERE Emp_Id = 122;
cqlsh:employee>
```

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (130, 'TempUser', 'Intern', '2024-01-01', 20000, 'CSE')
            ... USING TTL 15;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
cqlsh:employee>
```